CVD Prediction Project Report

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1. Background

Cardiovascular disease is the number one cause of death worldwide. It kills around 20.5 million people per year. The big problem is that CVD often develops silently, and by the time someone feels chest pain or shortness of breath, damage has already happened. Traditional clinical tools like the Framingham score have AUCs around 0.75–0.80, which leaves room for improvement. Our goal was to build machine learning models that can predict CVD earlier and more accurately, with a focus on recall since missing a real case (false negative) is the most dangerous outcome.

2. Dataset

We used the Sulianova Cardiovascular Disease dataset from Kaggle,which has 918 observations and 12 attributes, including age, sex, chest pain type, blood pressure, cholesterol, and max heart rate. It's one of the more commonly used heart disease datasets in research, though it's relatively small and turned out to be very linear, which ended up being the most important finding of our project.

3. Related Work

We looked at a few papers to understand what others have done. A PMC/NIH study evaluated multiple ML classifiers on similar structured heart datasets and found that ensemble methods consistently beat traditional scoring. An MDPI paper used basically the same Kaggle dataset and confirmed that preprocessing and feature selection heavily impact results. A broader literature review on ResearchGate showed no single model wins every time; it really depends on the data. We also looked at XGBoost benchmarks, where it hit 98% accuracy on a heart disease dataset with over 1,000 records, which is why we expected it to be our strongest model.

4. Preprocessing Method

All three models used the exact same preprocessing pipeline to keep comparisons fair. We replaced 172 zero values in the cholesterol column with the median, one-hot encoded 5 categorical columns (expanding features from 11 to 20), split the data 80/10/10, and standardized using only the training set to avoid leakage.

Models:

We each worked on and implemented three different models:

Teammate 2: Logistic Regression is our linear baseline. Fast, interpretable, and commonly used as a starting point in CVD research.

Sultan: XGBoost is a gradient boosted ensemble we expected to perform best based on the literature. Handles overfitting well and is the dominant method in tabular medical ML tasks.

Teammate 1: MLP is a small neural network we kept shallow on purpose to avoid vanishing gradients and overfitting, given our small dataset size.

5. Results

	Logistic Regression	XGBoost	MLP
Accuracy	86.96%	86.96%	84.78%
AUC	0.9231	0.9246	0.8720
Precision (Disease)	0.88	0.88	0.90
Recall (Disease)	0.91	0.91	0.84
F1 (Disease)	0.89	0.89	0.87

6. Conclusions

The main takeaway is that model complexity doesn't always mean better performance. On a small, linear dataset like this, a simple Logistic Regression is just as effective as XGBoost and a lot easier to interpret, which actually matters a lot in a clinical setting. If we had a much larger dataset, we'd expect XGBoost and especially the MLP to pull ahead.